

# Challenge Write-Up: Undo

## Challenge Information

|                     |                     |
|---------------------|---------------------|
| Student Name        | Keyzia Joy Pascua   |
| Section / Class     | BSIT 3 - Networking |
| Challenge Name      | Undo                |
| Category            | General Skills      |
| Points / Difficulty | Easy                |
| Date Solved         | August 11, 2026     |
| Flag Format Given   | picoCTF{...}        |

## 1. Challenge Description

Undo is a guided text-transformations challenge: the server has taken the original flag and applied a series of five Linux text transformations to it, one after another (encoding, reversing, character substitution, and so on). The task is not to find the flag but to recover it — at each step the server shows the currently transformed flag plus a hint describing what was just done to it, and the player has to supply the exact Linux command that undoes that one transformation, working backward until the original flag reappears.

## 2. Initial Analysis / Reconnaissance

The instance is a guided, interactive TCP service reached with nc rather than a static file to inspect, so there was nothing to enumerate up front — the service itself walks the player through each step and states exactly which transformation needs to be reversed. The only “analysis” needed was reading each hint carefully and mapping it to the correct standard Linux text-processing command (base64, rev, tr) with the right arguments, since the server checks the exact command text, not just whether the resulting flag would be correct.

### 3. Tools and Commands Used

| Tool / Command                                | What it was used for                                                                                          | Result / Output                                          |
|-----------------------------------------------|---------------------------------------------------------------------------------------------------------------|----------------------------------------------------------|
| <code>nc foggy-cliff.picoctf.net 56085</code> | Connect to the Text Transformations challenge service                                                         | Connected; server printed the challenge intro and Step 1 |
| <code>base64 -d</code>                        | Reverse a Base64 encoding applied to the flag string                                                          | Correct! advanced to Step 2                              |
| <code>rev</code>                              | Reverse a full string reversal applied to the flag                                                            | Correct! advanced to Step 3                              |
| <code>tr '-' '_'</code>                       | Reverse a substitution that had replaced underscores with dashes, by mapping dashes back to underscores       | Correct! advanced to Step 4                              |
| <code>tr '()' '{}'</code>                     | Reverse a substitution that had replaced curly braces with parentheses, by mapping parentheses back to braces | Correct! advanced to Step 5                              |
| <code>tr 'A-Za-z' 'N-ZA-Mn-za-m'</code>       | Reverse a ROT13 substitution applied to the letters of the flag                                               | Correct! server printed the recovered original flag      |

### 4. Solution Walkthrough

#### Step 1

Connect with `nc foggy-cliff.picoctf.net 56085`. The service explains the rules: at each step it shows a transformed version of the flag plus a hint about what was just done to it, and expects the exact Linux command that would undo that transformation. Step 1's hint says “Base64 encoded the string”, so the correct undo command is `base64 -d`, which decodes Base64 back to plain text. The server accepts it and moves to Step 2.

#### Step 2

Step 2's hint says “Reversed the text” — the whole string was flipped end-to-end (the flag output at this stage visibly starts with the closing brace and ends near the picoCTF{ opener, backwards). The command that reverses a reversed string is simply `rev`, which reverses the order of characters on a line. The server accepts it and moves to Step 3.

#### Step 3

Step 3's hint says “Replaced underscores with dashes”, so every original `_` in the flag has become a `-`. To undo that character-for-character substitution, `tr` translates dashes back to underscores: `tr '-' '_'`. The server accepts it and moves to Step 4.

#### Additional steps (add as needed)

Step 4's hint says “Replaced curly braces with parentheses”, so the flag's `{` and `}` delimiters had become `(` and `)`. Reversing that substitution with `tr '()' '{}'` restores the braces and advances to Step 5. Step 5's hint says “Applied ROT13 to letters”, a classic 13-position alphabet rotation; reversing ROT13 is done with the same rotation run again in the opposite direction, expressed as `tr 'A-Za-z' 'N-ZA-Mn-za-m'`. After this final correct command, the server congratulates the player and prints the fully recovered original flag: `picoCTF{Revers1ng_t3xt_Tr4nsf0rm@t10ns_aa00e8fb}`.

```
>_ Terminal  CyberChef  CTF Primer  →

^C
pascuakeyzia-academy@webshell:~$ nc foggy-cliff.picoctf.net 56085
===Welcome to the Text Transformations Challenge!===

Your goal: step by step, recover the original flag.
At each step, you'll see the transformed flag and a hint.
Enter the correct Linux command to reverse the last transformation.

--- Step 1 ---
Current flag: KW9z0HIwMG5uLWZhMDFnQHplMHNmYTRlRy1nazNnLXRhMWZlcmlyRShTR1BicHZj
Hint: Base64 encoded the string.
Enter the Linux command to reverse it: base64 -d
Correct!

--- Step 2 ---
Current flag: )os8r00nn-fa01g@ze0sfa4eG-gk3g-ta1ferirE(SGPbpvc
Hint: Reversed the text.
Enter the Linux command to reverse it: rev
Correct!

--- Step 3 ---
Current flag: cvpbPGS(Eriref1at-g3kg-Ge4afs0ez@g10af-nn00r8so)
Hint: Replaced underscores with dashes.
Enter the Linux command to reverse it: tr '-' '_'
Correct!

--- Step 4 ---
Current flag: cvpbPGS(Eriref1at_g3kg_Ge4afs0ez@g10af_nn00r8so)
Hint: Replaced curly braces with parentheses.
Enter the Linux command to reverse it: tr '()' '{} '
Correct!

--- Step 5 ---
Current flag: cvpbPGS{Eriref1at_g3kg_Ge4afs0ez@g10af_nn00r8so}
Hint: Applied ROT13 to letters.
Enter the Linux command to reverse it: tr 'A-Za-z' 'N-ZA-Mn-za-m'
Correct!

Congratulations! You've recovered the original flag:
>>> picoCTF{Revers1ng_t3xt_Tr4nsf0rm@t10ns_aa00e8fb}

```

Terminal session showing all five reverse-transformation steps and the recovered flag.

## 5. Flag

```
picoCTF{Revers1ng_t3xt_Tr4nsf0rm@t10ns_aa00e8fb}
```

## 6. Key Concept / What I Learned

The key concept is working backward through a chain of standard Linux text transformations by recognizing each one from its effect and knowing (or being reminded of) its inverse: base64/base64 -d for encoding, rev for a full-string reversal, tr for one-to-one character substitution (including ROT13, which is simply tr with the alphabet rotated 13 places and is its own perfect inverse when reapplied). It reinforces that tr's two

arguments are a character-by-character mapping, so undoing a substitution just means swapping the order of the “from” and “to” sets.

## **7. References**

None beyond the challenge's own step-by-step hints and standard documentation for `base64`, `rev`, and `tr`.